ISERS: Intelligent Staff Rostering System for Hospitality Workforce Management

Automated Staff Rostering System for Infinity Hospitality Limited

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1. Background & Motivation

- Hospitality rostering at Infinity Hospitality relied on spreadsheets, calls, and messaging.
- Fragmented data caused slow scheduling, weak traceability, and difficult replacements.
- A centralised, explainable, and location-aware workforce platform was needed.

2. Objectives

- Centralise employee, skill, leave, and staffing requirement data.
- Generate explainable rosters using hard and soft constraints.
- Support manager override, automated replacement, employee self-service, and reporting.
- Improve transparency, compliance, and workforce intelligence.

3. System Design / Workflow

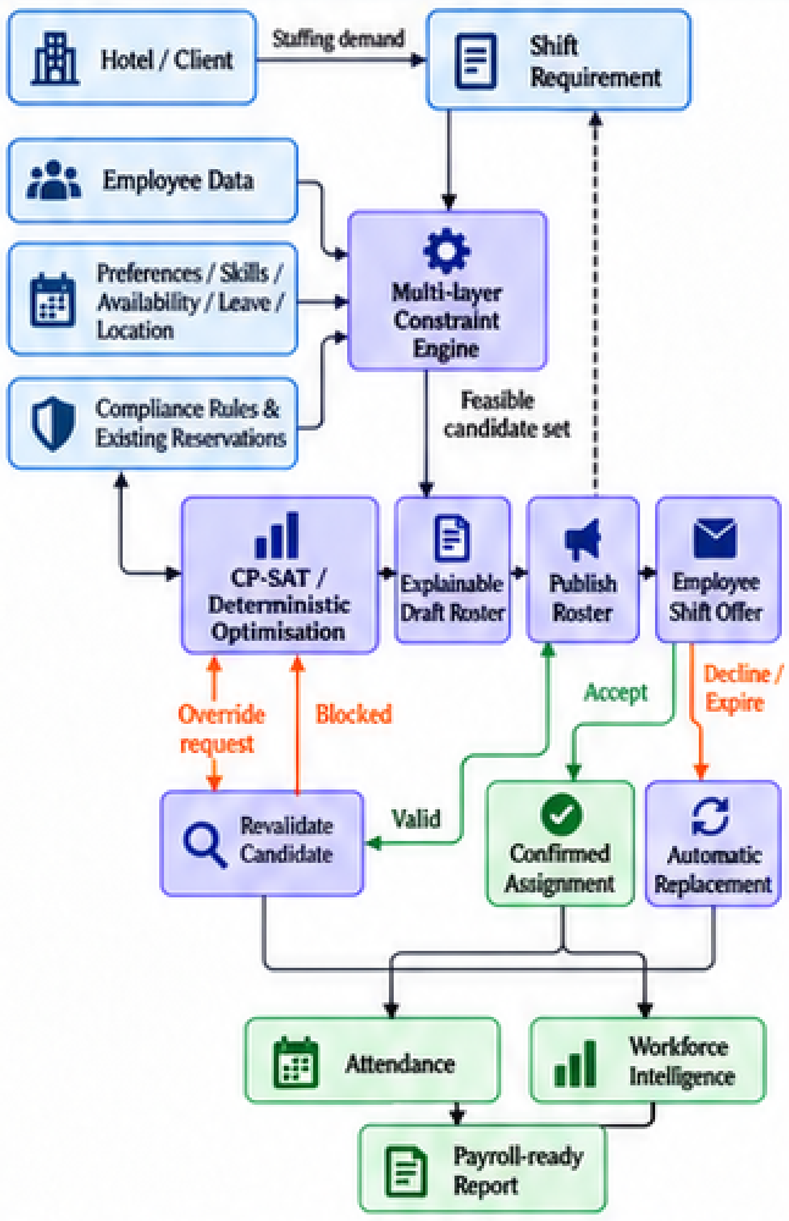


Figure 1. ISERS workflow and decision-support process.

- Role-aware flow for managers and employees, from demand to roster publication and shift offers.
- Candidate eligibility pipeline with constraint-based optimisation and explainable results.
- Human-in-the-loop manager control with override, validation and automated replacement.

4. Technical Architecture

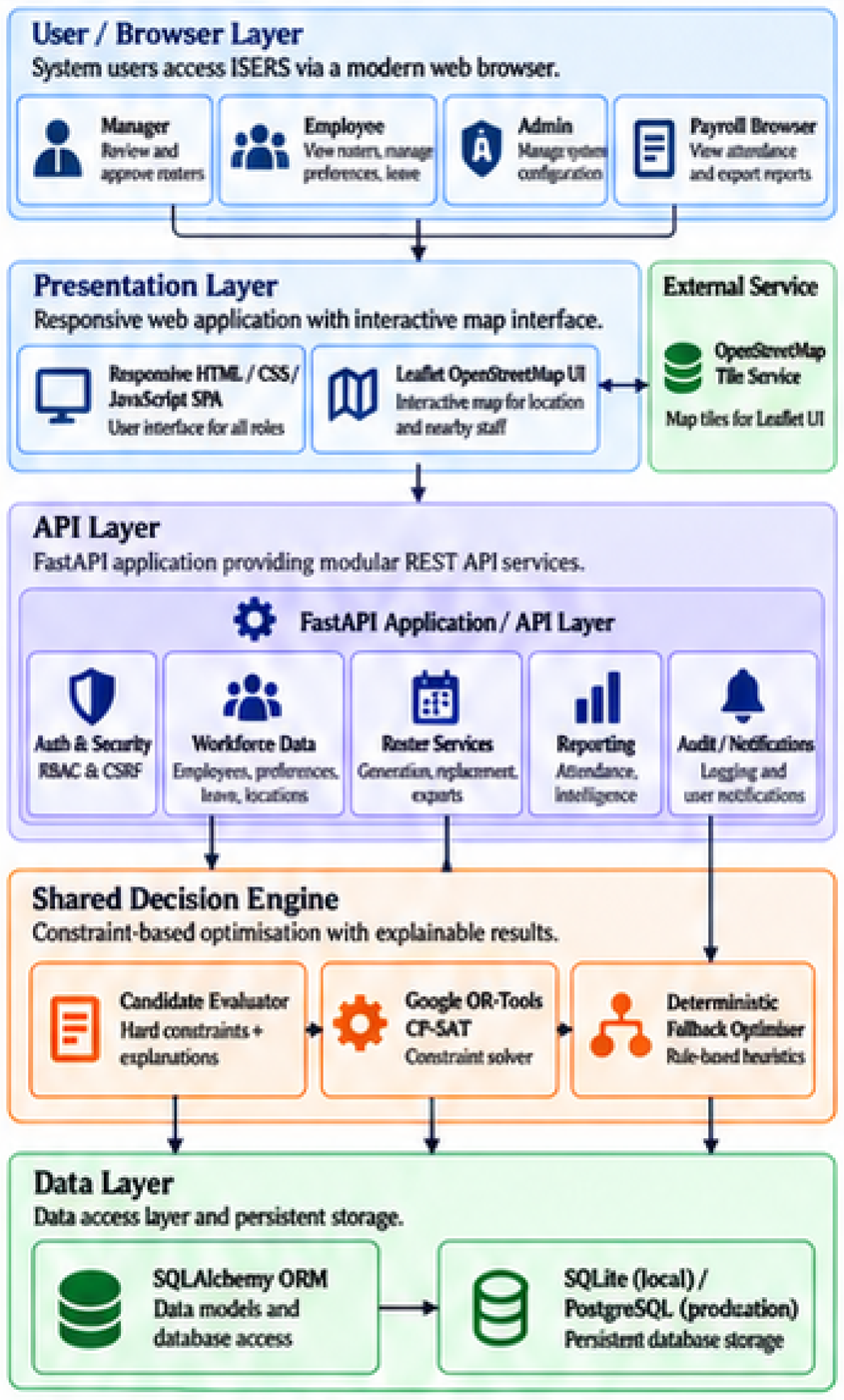


Figure 2. Final technical architecture of ISERS.

- Responsive HTML/CSS/JavaScript frontend with Leaflet + OpenStreetMap.
- FastAPI backend with modular REST APIs.
- SQLAlchemy ORM with SQLite for local/demo and PostgreSQL for deployment.
- Google OR-Tools CP-SAT with deterministic fallback.
- Render-based deployment.

5. Core Functional Modules

- Manager Command Centre
- Employee & skill management
- Staffing requirements
- Automated roster generation
- Explainable candidate explorer
- Map & nearest suitable worker
- Employee self-service portal
- Offer / replacement workflow
- Compliance rules
- Attendance, audit and payroll-ready reporting
- Workforce intelligence and what-if analysis

6. Demonstration Screenshots

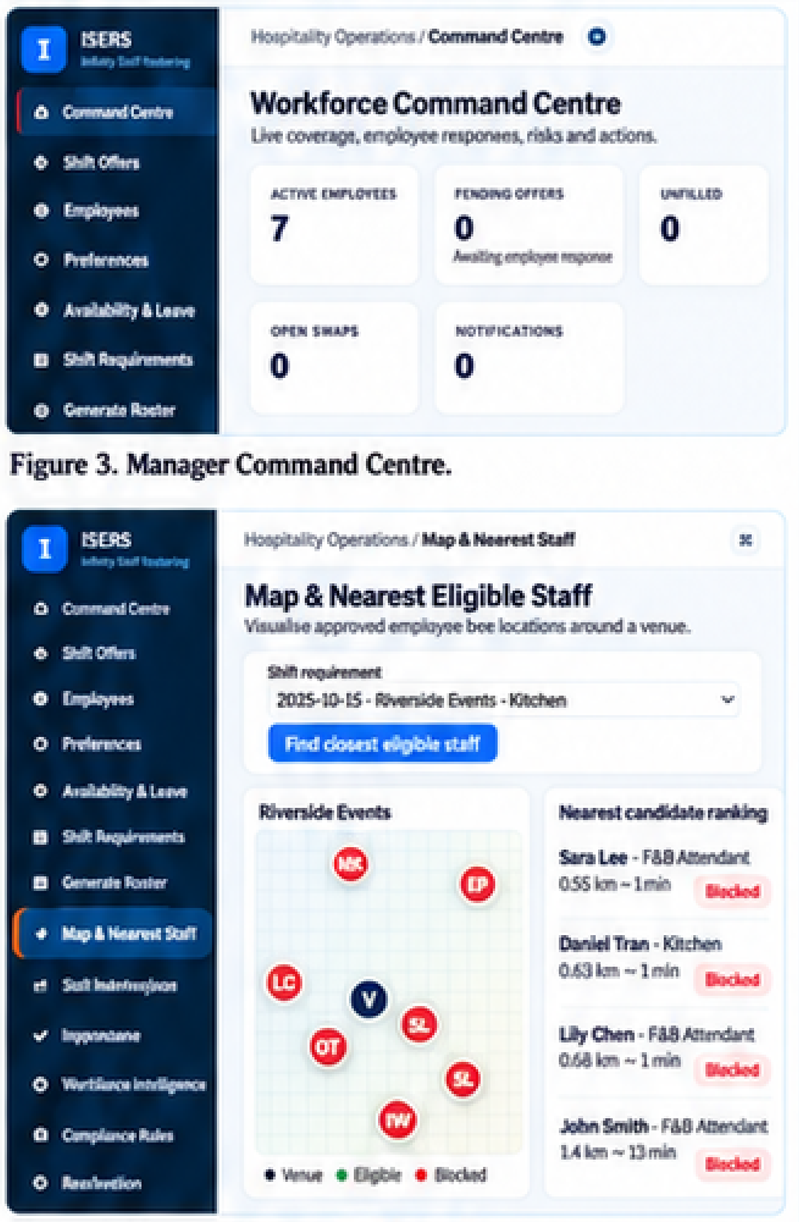


Figure 3. Manager Command Centre.

Figure 4. Map and nearest eligible staff.

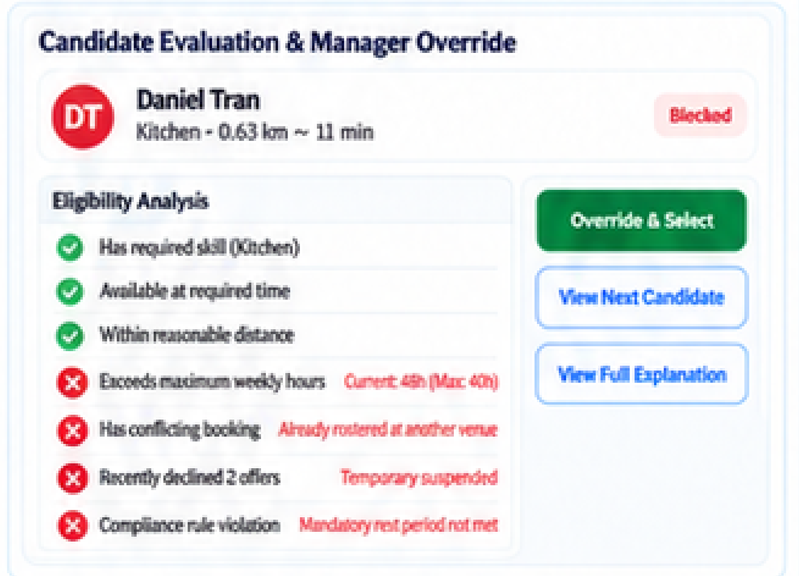
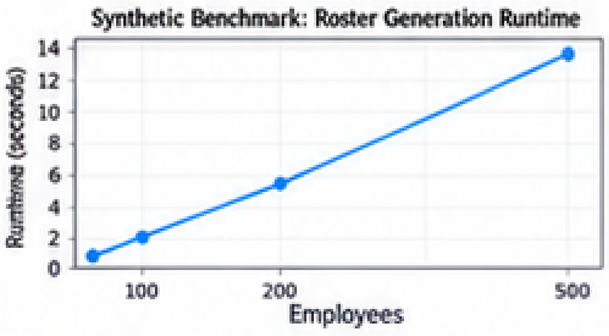


Figure 5. Explainable candidate evaluation and manager override.

7. Key Innovation / Highlights

- Explainable candidate blocking reasons.
- Multi-layer hard and soft constraints.
- Automated next-candidate replacement.
- Two-consecutive non-acceptance suspension and reactivation workflow.
- Cross-manager conflict protection.
- Location-aware candidate comparison.

8. Results & Evaluation



Employees	Positions Filled	Coverage	Runtime
50	30	100%	1.2955 s
100	60	100%	2.530 s
250	150	100%	6.3657 s
500	300	100%	13.3572 s

- Automated test suite: 5/5 passed.
- Python compilation passed for 21 files.
- JavaScript syntax and CSS parsing checks passed.
- Controlled synthetic tests achieved 100% staffing across all cases.

Figure 6. Synthetic benchmark: roster generation runtime.

9. Pilot Validation

Context: Housekeeping Appreciation Week.

Participants: approximately 15 employees.

Outcome: improved visibility, structured allocation, and positive stakeholder feedback.

Results indicate technical feasibility and practical value, not full commercial-scale validation.

10. Conclusion & Future Work

ISERS integrates optimisation, explainability, location-awareness, manager oversight, and employee participation in one platform.

Future work: deeper payroll integrations, broader field trials, extended analytics, and long-term forecasting.

11. References

[1] S. Bisterzilli and M. Mohc, "Solving the general employee scheduling problem," *Computers & Operations Research*, 2018.
[2] L. Pinetti, R. Gaber, and S. Ore, "The OR-Tools CP solver," CP 2013.
[3] Google, "Employee scheduling with OR-Tools," 2016.
[4] K. Gunocrason et al., "Explainable AI for enhanced decision-making," *Decision Support Systems*, 2020.

12. Live Demo

Live Demo

Scan to explore the live deployed system.

https://qrlify.io/Gu7EΓHTWq